

KonthoProhori

Bangla Deepfake Audio Detection and Voice Forensics

Track E - National Defence | SciBlitz AI Challenge 2026

KonthoProhori is a privacy-aware web application for screening suspicious Bangla voice clips. It combines exact notebook-compatible acoustic feature extraction, a saved StandardScaler, a trained XGBoost classifier, bilingual risk communication, and downloadable analysis reports.

Item	Value
Live demo	TBD - update after final deployment
Repository	GitHub repository on branch/PR
Main reported result	99.57% accuracy on notebook random held-out split
Important caveat	Cross-speaker, cross-generator, and real-world validation remains future work

1. Problem Statement

AI-generated Bangla speech can imitate public officials, security personnel, journalists, executives, relatives, and ordinary citizens. Suspicious voice messages can spread quickly through social media and messaging platforms, creating risk for misinformation, scams, fake emergency announcements, reputational harm, and public-trust manipulation.

Bangla-speaking communities have limited access to locally focused tools for quickly screening suspicious audio. KonthoProhori addresses this gap as a decision-support system, not as standalone legal or forensic proof.

2. Proposed Solution

The product allows a user to upload an audio file or record through the browser, then receive a three-way screening result: Likely Genuine, Inconclusive, or Likely Deepfake. The interface displays model probabilities, waveform, log-mel spectrogram, grouped acoustic evidence, limitations, and a downloadable JSON report.

Target Users

- Journalists and fact-checkers who need fast screening before publishing.
- Public communication teams and cybercrime units handling suspicious clips.
- Financial institutions, call centers, moderators, and Bangla-speaking citizens.

3. Methodology

The backend validates input, decodes audio, converts it to mono, resamples to 16 kHz, and analyzes the first five seconds. Invalid, empty, corrupt, unsupported, oversized, too-short, and near-silent files are rejected with safe user messages.

Step	Implementation
Decode	SoundFile primary path with Librosa fallback where needed
Standardization	Mono audio, 16 kHz sample rate
Window	First 5.0 seconds, matching the notebook
Validation	Duration, finite values, size, extension, RMS, peak level
Privacy	No raw audio in report and no intentional retention

Feature Engineering

Index range	Feature family	Dimensions
0:40	MFCC mean	40
40:80	MFCC standard deviation	40
80:92	Chroma mean	12
92:220	Mel-spectrogram mean	128
220:227	Spectral-contrast mean	7
Total		227

4. AI/ML Approach

The classifier is the saved XGBoost binary model from the existing notebook. The application does not retrain or modify the model assets. The saved StandardScaler is applied before inference. The model returns probabilities for real and fake classes, which are communicated as model probabilities.

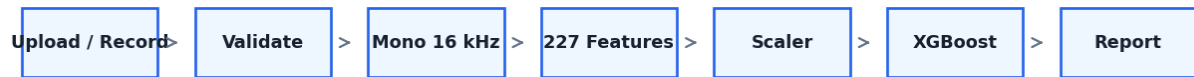
Fake probability	Product decision
≥ 0.70	Likely Deepfake
≤ 0.30	Likely Genuine
Between 0.30 and 0.70	Inconclusive

These are heuristic product-level decision bands for uncertainty-aware communication. They are not newly trained thresholds.

5. Architecture And Implementation

The system uses a FastAPI backend and a Next.js frontend. The frontend handles audio selection, microphone recording, loading states, results, charts, and report download. The backend owns all ML execution and returns structured JSON.

KonthoProhori inference flow



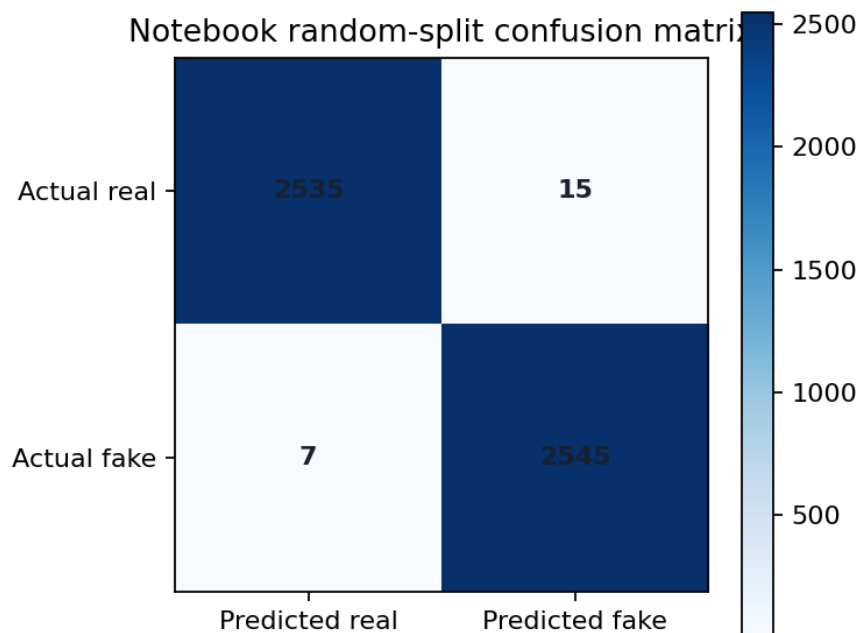
Module	Role
frontend/	User workflow and browser microphone capture
backend/main.py	HTTP API, CORS, safe error conversion
src/audio_io.py	Validation, decode, mono conversion, resampling
src/features.py	Exact 227-feature extraction
src/predictor.py	Cached scaler/model loading and prediction
src/visualization.py	Waveform, spectrogram, probability, evidence charts
src/reporting.py	Downloadable JSON result

Deployment

The recommended deployment is Railway for the FastAPI backend and Vercel for the Next.js frontend. The frontend uses NEXT_PUBLIC_API_BASE_URL to call the Railway backend. Railway serves /api/health and /api/analyze.

6. Results

The existing notebook reports 99.57% accuracy on a random stratified held-out split. The confusion matrix is shown below. This metric is useful evidence for the available dataset but must not be presented as universal field accuracy.



Metric	Value
Evaluation type	Random stratified hold-out
Test size	5,102
Accuracy	99.57%
Real support	2,550
Fake support	2,552

Validation Caveat

Broader cross-speaker, cross-generator, and real-world validation remains future work. Possible risks include speaker overlap, recording-condition overlap, generator-specific artifacts, compression, background noise, phone audio, and unseen synthesis systems.

7. Impact, Privacy, And Ethics

KonthoProhori supports national-security and public-information integrity by helping users screen suspicious Bangla voice clips before trusting, forwarding, publishing, or acting on them.

Responsible Use Controls

- The tool provides AI-assisted screening, not legal or forensic proof.
- The app does not identify who is speaking.
- Uploaded audio is not intentionally retained or used for retraining.
- Reports exclude raw audio and absolute server paths.
- The UI encourages source verification and expert review for high-risk cases.

Third-Party Attribution

Dataset reference: Bangla DeepFake Dataset by ahanaf101 on Kaggle. Libraries include FastAPI, Uvicorn, Librosa, SoundFile, NumPy, Scikit-learn, XGBoost, Joblib, Matplotlib, Next.js, React, and Lucide React.

8. Limitations And Future Work

- Confirm dataset license from Kaggle before final submission.
- Add consented real and synthetic examples if redistribution is permitted.
- Evaluate on an external challenge set with multiple speakers and devices.
- Run robustness tests for MP3 compression, noise, phone quality, and low volume.
- Perform speaker-disjoint and cross-generator evaluation if metadata allows.
- Add segment-level analysis for longer clips as an experimental feature.

Conclusion

KonthoProhori turns the existing Bangla deepfake audio model into a usable, privacy-conscious, and uncertainty-aware screening product. It is designed to support human verification and professional review, not replace them.